Forecasting Migraine Attacks

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Disclosures

- Statistical Editor, *Anesthesiology*
- Statistical Editor, *Headache*

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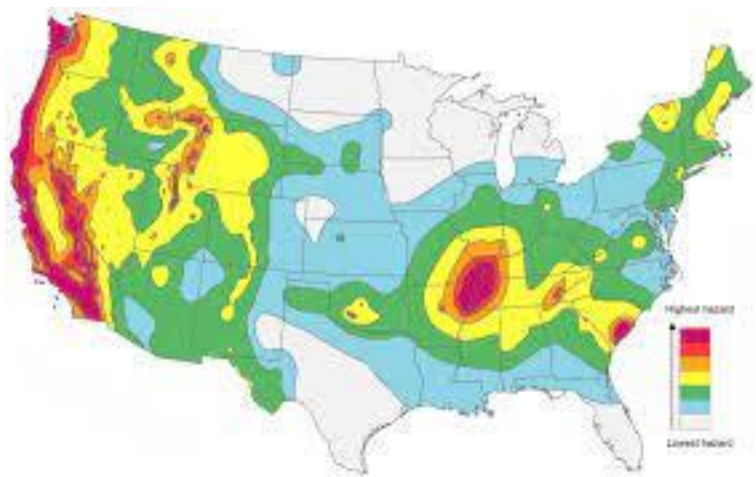
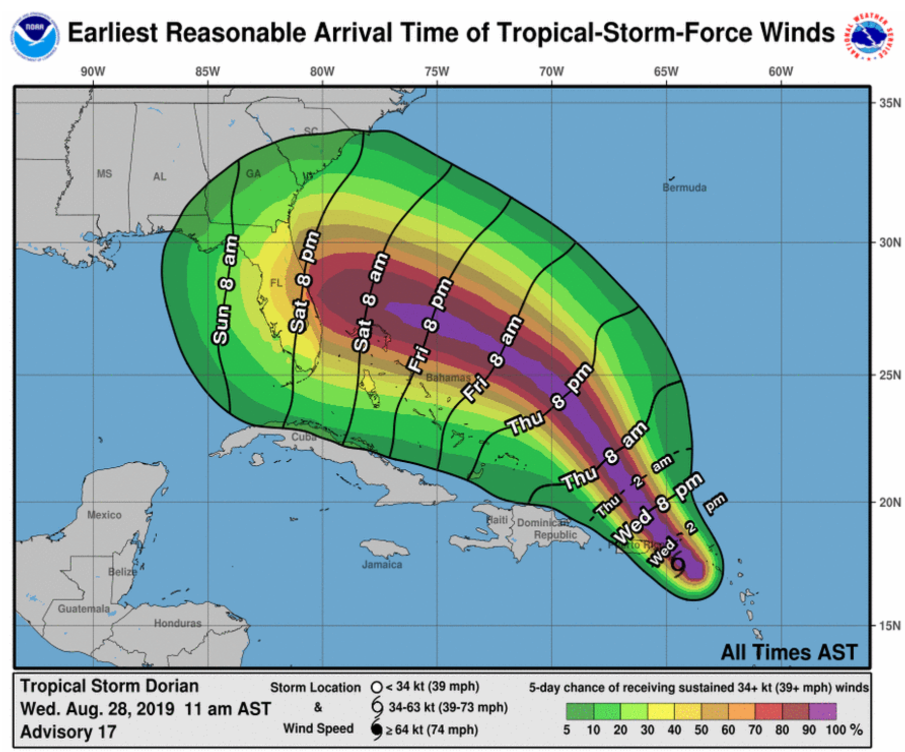
- StatReviewer Software
  - Co-founder

Can Migraine Attacks be Predicted?

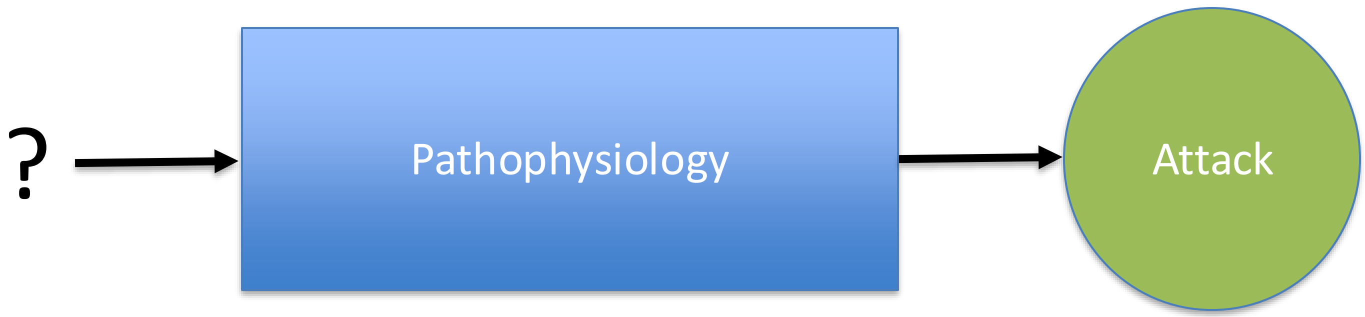


Earthquake Forecast



Hurricane Forecast

What Causes Migraine Attacks?



- Vascular theory
- Cortical Spreading Depression
- Trigeminovascular Pathway

Hippocrates (300 BC)

Behavioral	External
Foods Exercise Hunting	Winds and weather
	Internal Gases in blood Thirst (dehydration) Poorly digested foods

Galen (150 AD)

Behavioral	External
Wine Spices (mustard etc…) Errors in diet Smelling Egyptian incense Vinegar	Heat Cold
	Internal 4 humors Menstrual cycle

Avicenna(1000 AD)

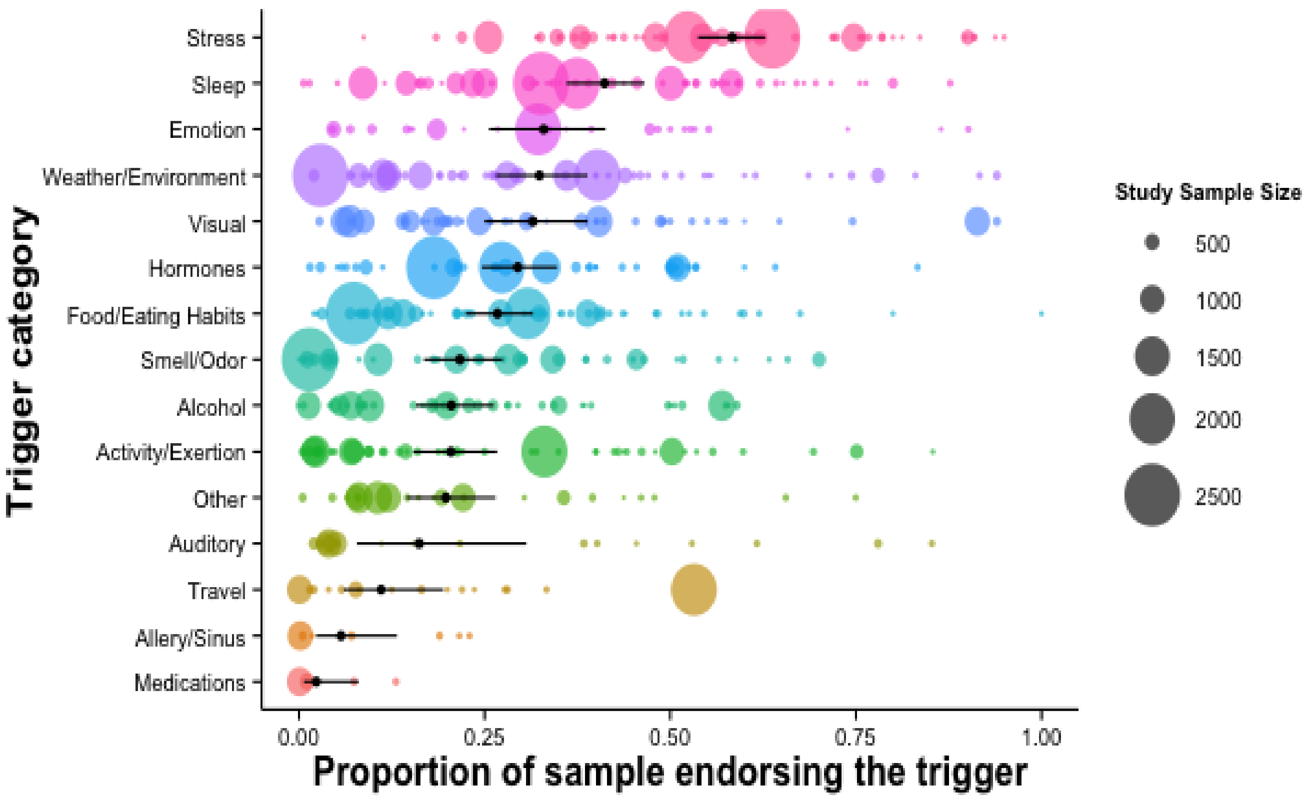
Behavioral	External
Foods Gluttony	Weather Celestial events
	Internal Humoral theory Organ systems Depression Hormones

Willis (1650 AD)

Behavioral	External
Error in diet Drunkenness Bathing Reading	Cold weather Precipitation Change in weather Equinox and solstices
	Internal Vascular theory

Headache Trigger Meta-Analysis

Figure 3.



Walters-Pelligrino et al, Cephalagia, 2017

What/how do patients come to know about headache triggers?

- Knowledge about triggers is not inherent so these factors must be learned
  - Media
  - Practitioners
  - Experience

Headache Sufferers Develop Causal Models to Predict Their Attacks:

- 85% of sufferers report at least 1 trigger (Paulin et al., 1987)
- Sufferers report a median of 3 - 7 triggers (range 0 – 12)
- When prompted, 95% of sufferers will report at least one trigger

(Blau, 1988; Van den Bergh, 1987; Kelman, 2007)

What/How do we know about the potency of headache triggers?

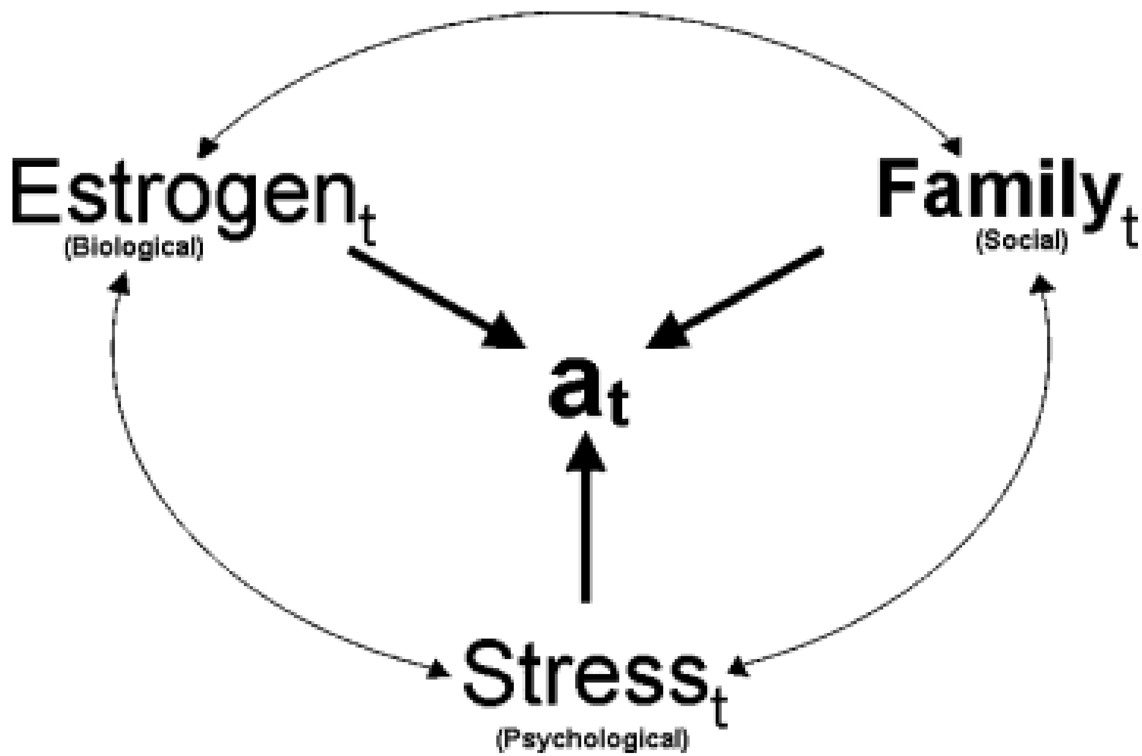
- The overwhelming majority of studies are conducted using survey research and self-report
- Sufferers are either asked to report what “causes” their headaches in an open-ended question, or given lists of potential triggers

RCTs have been mixed

- Chocolate (Marcus et al, 1997) ✕
- Stress (Gannon, Haynes, et al 1987, 1990)
- Aspartame (Schiffman, 1987) ✕
- Mood (Martin et al, 1997, 1999, 2005)
- Food antigens (Alpay et al, 2010) ~

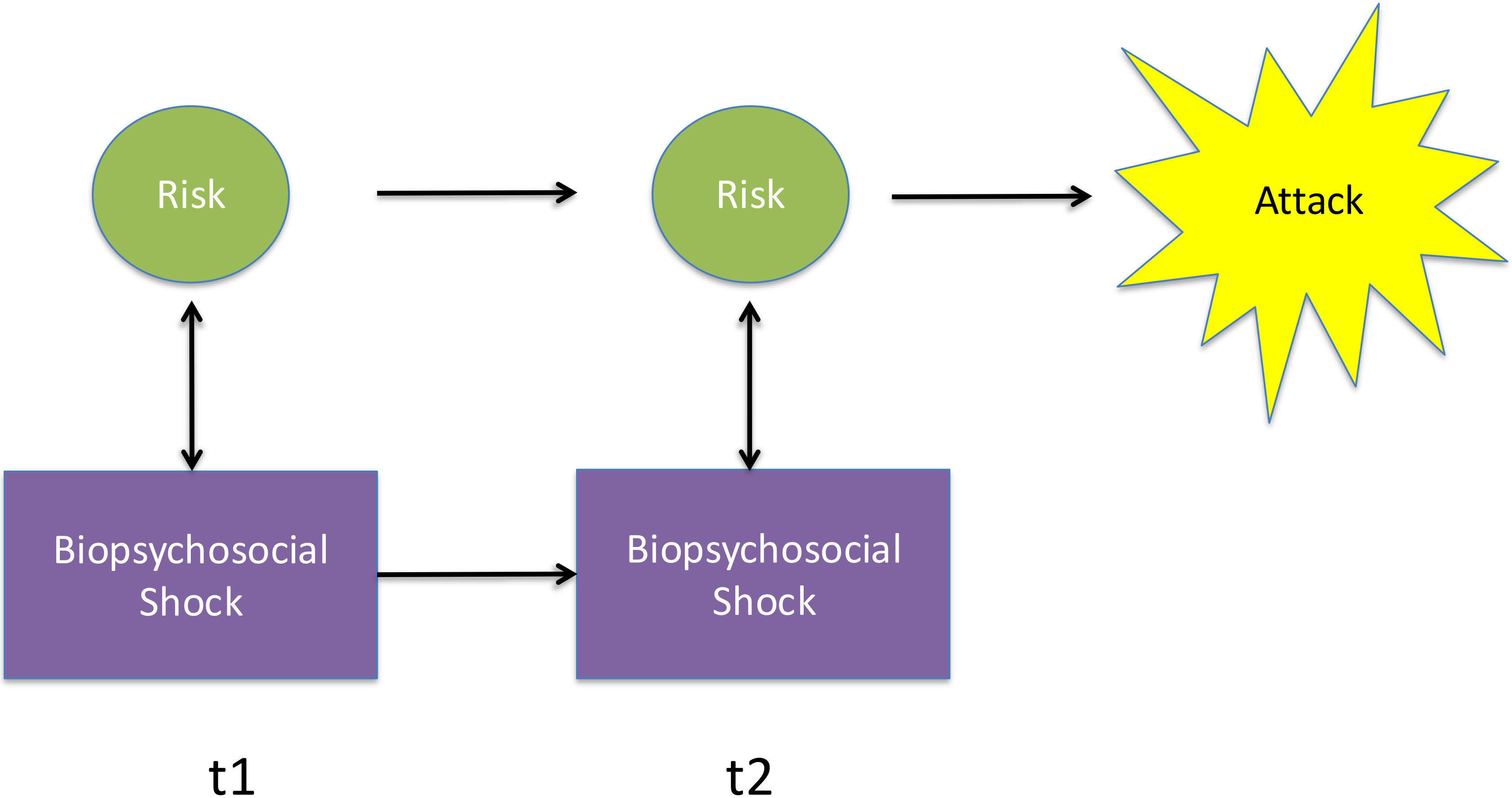
What To Include in a Headache Forecasting Model?

A Forecasting Treatment Paradigm: Biopsychosocial Process Model



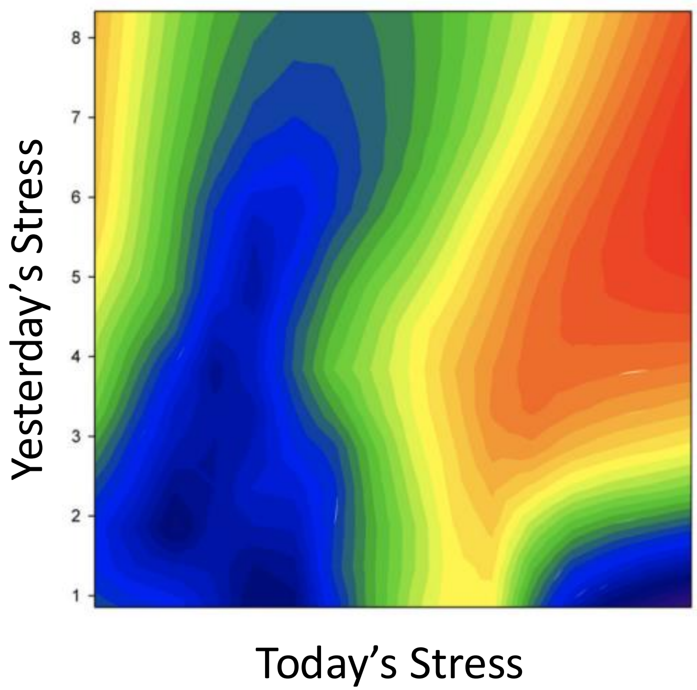
Houle TT, Remble TA, Houle TA. The examination of headache activity using time series research designs. Headache J Head Face Pain. 2005;45:438-444.

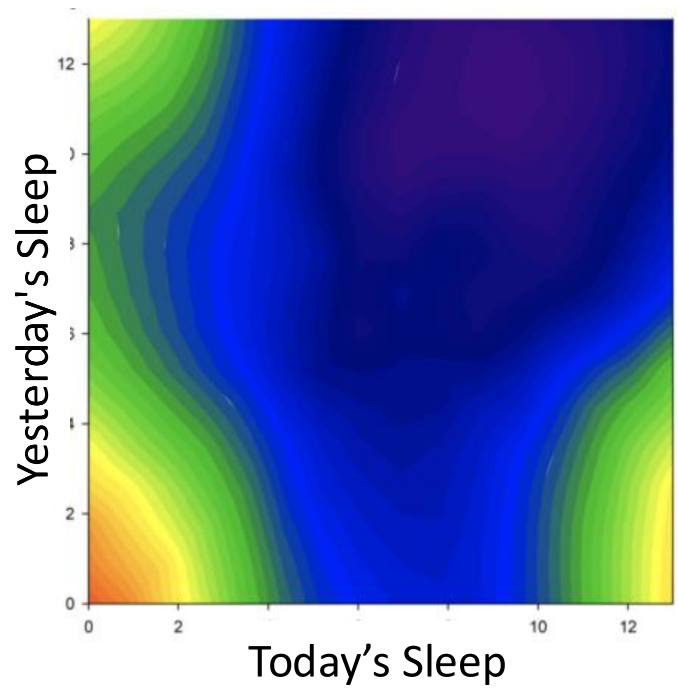
A Forecasting Treatment Paradigm



Stress and Sleep Duration Predict Headache Severity in Chronic Headache Sufferers

- N = 55 Individuals with chronic headache
  - n = 22 Tension-Type headache
  - n = 33 Migraine
- Headache frequency 26 (15 to 28)





Houle et al., 2012, PAIN

Headache Prediction Study (HAPRED)

- Prospective longitudinal cohort study of episodic migraine sufferers Winston-Salem, NC region from September 2009 to May 2014.

- Participants were recruited from the general community, neurology practices, and primary care clinics using direct television and print (e.g., fliers) advertising.

- Inclusion criteria:
  - either gender
  - episodic migraine with or without aura (IHS-II 2.1 or 2.2)
  - > 2 headache attacks/month (4 and 14 headache days/month)

Headache Prediction Study (HAPRED)

- Exclusion criteria:
  - the presence of a secondary headache disorder (e.g., brain tumor)
  - a recent change in nature of headache symptoms over last 6 weeks,
  - not being able to read or speak English at a 6th grade level
  - having a psychiatric hospitalization within the previous 12 months
  - meeting criteria for, or currently being treated for a DSM-IV diagnosis of substance abuse (not including AA, NA, etc.)
  - pregnancy or planned pregnancy during the observation period
  - lack of stable residence or contact number
  - meeting IHS-II criteria for medication overuse (greater than two days/week of abortive medication use for headache)

- Headache diagnosis and other medical exclusions were confirmed by a Neurologist using the structured diagnostic interview

Daily Assessment of Stress & Headache

- Participants completed two electronic diary entries each day (i.e., AM and PM).

Daily Assessment

- Headache Activity
- Sleep
- Stress
- Mood
- Diet (caffeine, alcohol, missed meals)
- Common triggers (certain foods, menstrual cycle)

Daily Stress Inventory (DSI)
Brantley et al, 1987

- 58 item scale that allows a subject to identify stressful events (i.e. daily hassles) that they have experienced in the last 24 hours

- "Interrupted during task/activity"
- "Waited longer than you wanted"

- Occurred (yes or no, rated 1-7)
  - 1 "occurred, but not stressful"
  - 7 "caused me to panic"

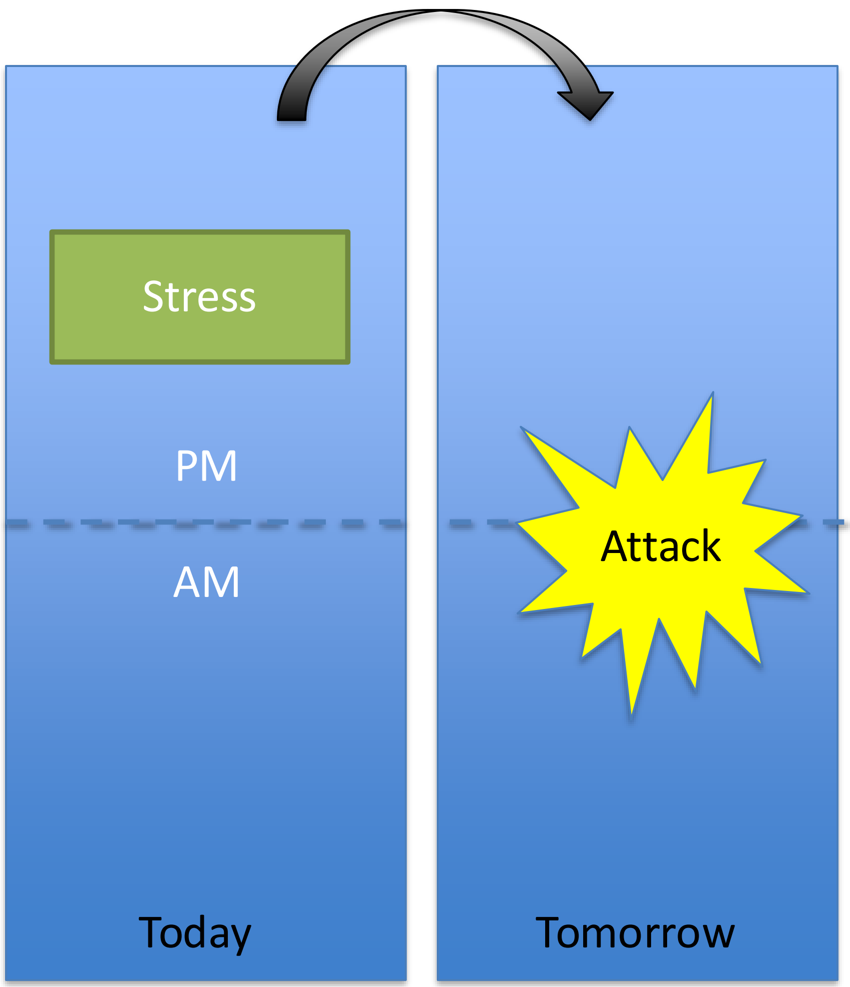
Subscales

Frequency (0 to 58)
Sum (0 to 406)
Average Intensity Rating (0 to 7)

Primary Outcome

- Prediction (forecast) of a future headache attack based on current levels of stress and headache.

- The primary outcome is the presence/absence of any headache attack (head pain > 0 on a NRS) occurring over the 24 hour period recorded following a PM diary entry.



Results

- N = 95/100 (95%) participants completed the study
  - 5 drop-out (contributed negligible data)
- 4626 diary days were completed
  - Mean 49 days (range: 11 – 96)
- 431 (9.3%) embedded missing diary days

Participant Characteristics	N = 95
Age	40.3 (12.9)
Gender	
Female	86 (90.5%)
Male	9 (9.5%)
Race/ethnicity	
Caucasian	83 (87.4%)
African American	9 (9.5%)
Other	3 (3.1%)
Diagnosis	
Migraine without aura	79 (83.2%)
Migraine with aura	10 (10.5%)
Migraine with TTH	6 (6.3%)
MIDAS	10 [3.25, 18]
CES-D	9 [4, 17.5]
STAI	36 [32, 43.5]

Mean (SD), Frequency (%), Median [25^{th}, 75^{th}]

Day-level Characteristics:

	No Headache Tomorrow	Headache Tomorrow	P-value
N	2241	1437	
Stress			
FREQ	5 [2, 9]	6 [3, 11]	<0.001
SUM	11 [4, 21]	14 [7, 27]	<0.001
AIR	2 [1.29, 2.94]	2.4 [1.7, 3.0]	<0.001
Headache Today	765 (34.1%)	698 (48.6%)	<0.001

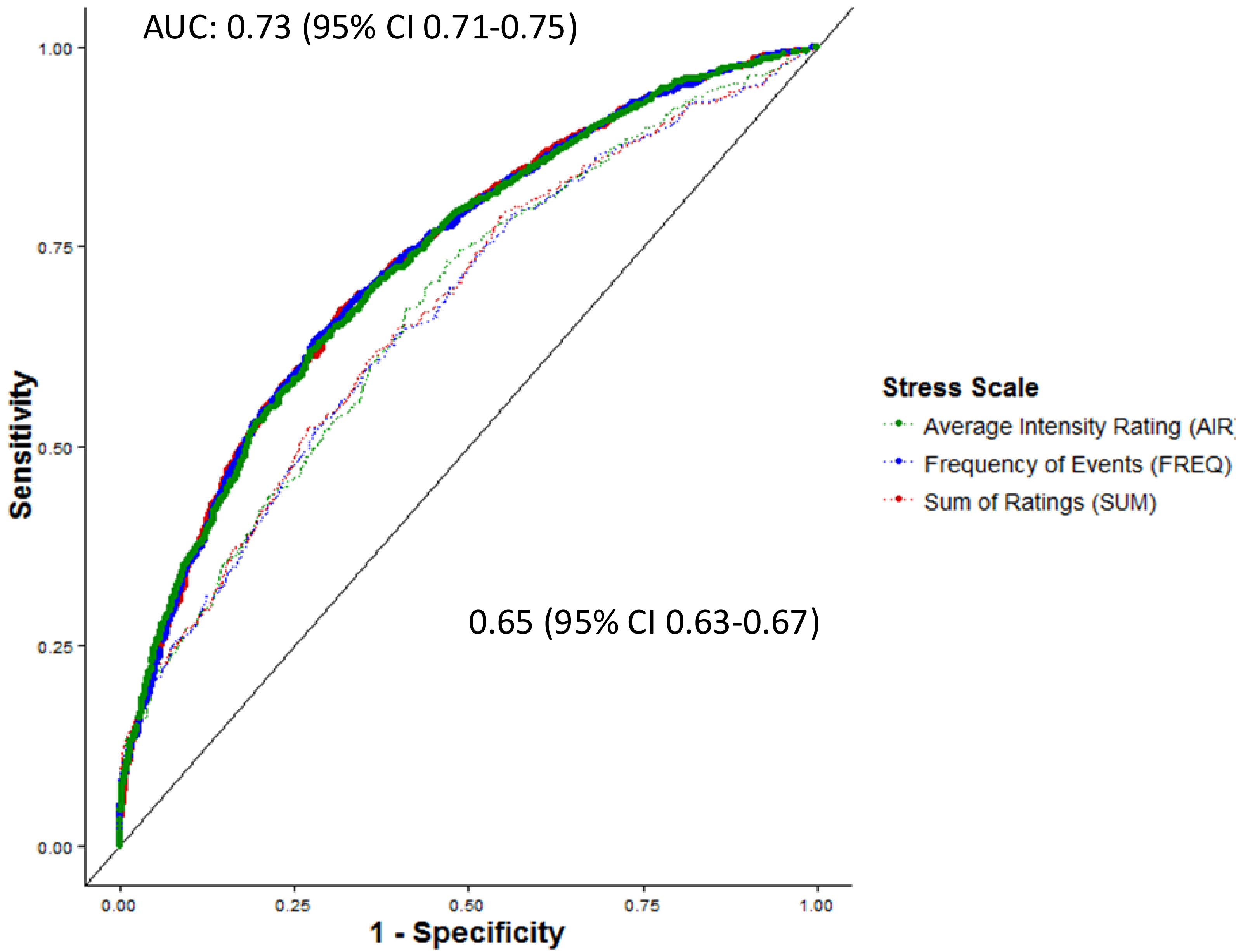
Headache occurred on: 1613/4195 (38.5%) days

Prediction Model

- Today's headache was associated with 45% increase in the odds of headache tomorrow
  - OR 1.45 (95%CI: 1.15 to 1.82), p = 0.0018

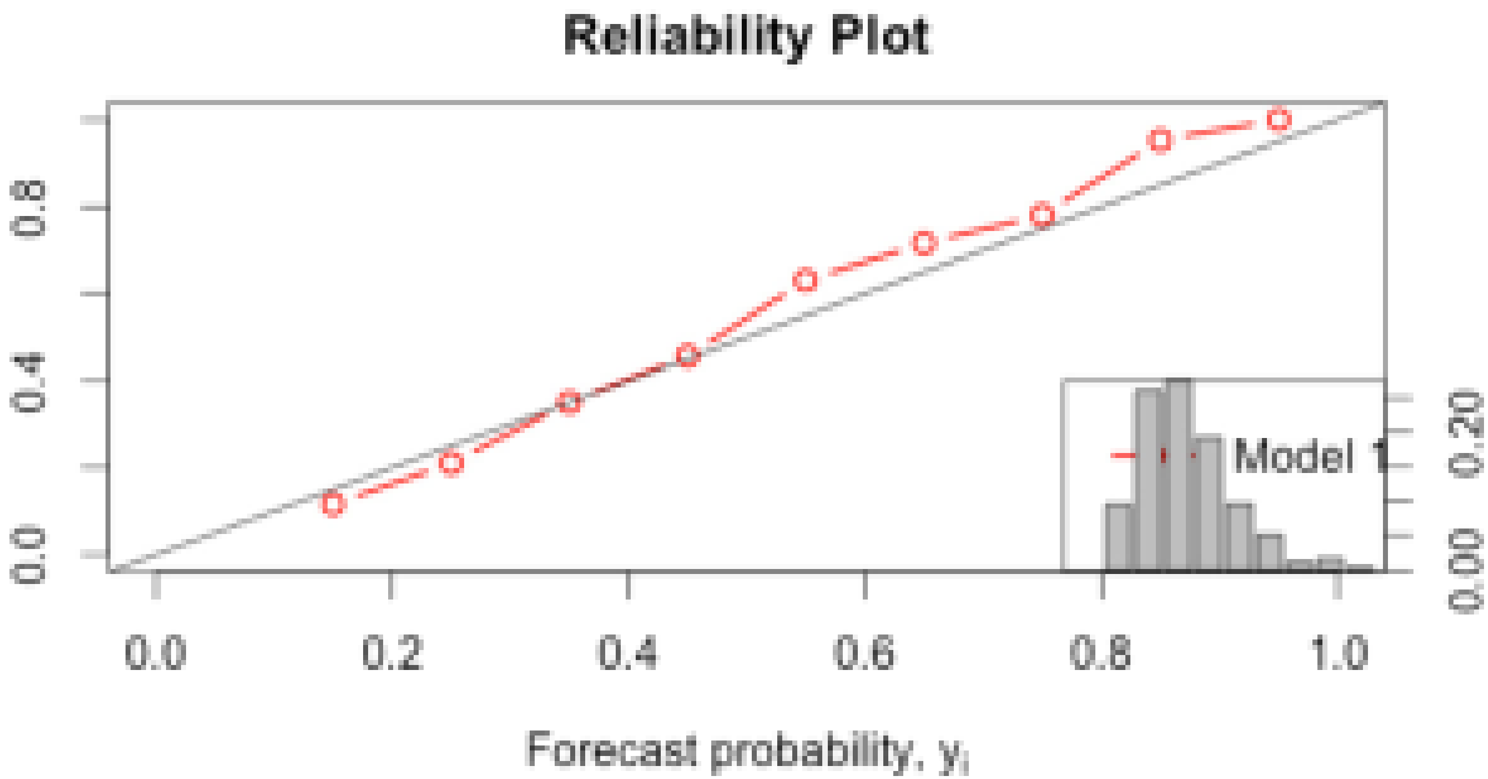
- Each point in today's stress (AIR) was associated with an 23% increase in the odds of headache tomorrow
  - OR 1.23 (95%CI: 1.12 to 1.34), p = <0.0001

Houle et al., 2017, Headache



AUC: 0.73 (95% CI 0.71-0.75)
0.65 (95% CI 0.63-0.67)
Sensitivity
1 - Specificity
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
Stress Scale
Average Intensity Rating (AIR)
Frequency of Events (FREQ)
Sum of Ratings (SUM)

Calibration & Resolution

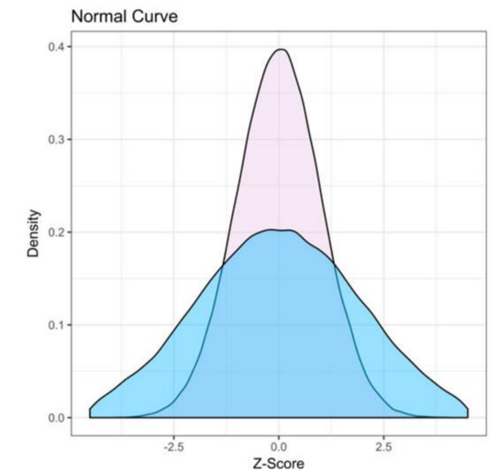


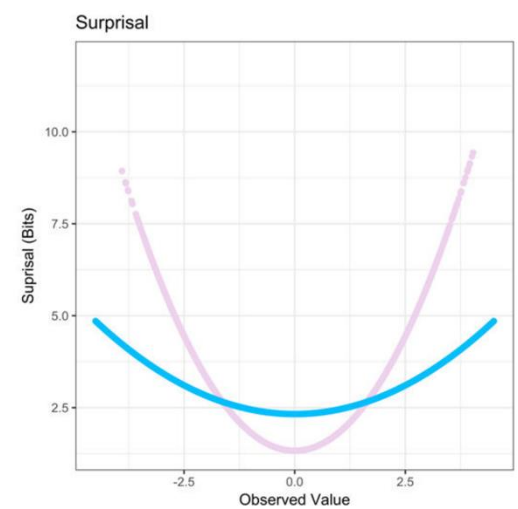
Headache Triggers as Surprise

Dana P. Turner MSPH, PhD, Adriana D. Lebowitz BA, Ivana Chtay BS, Timothy T. Houle PhD,

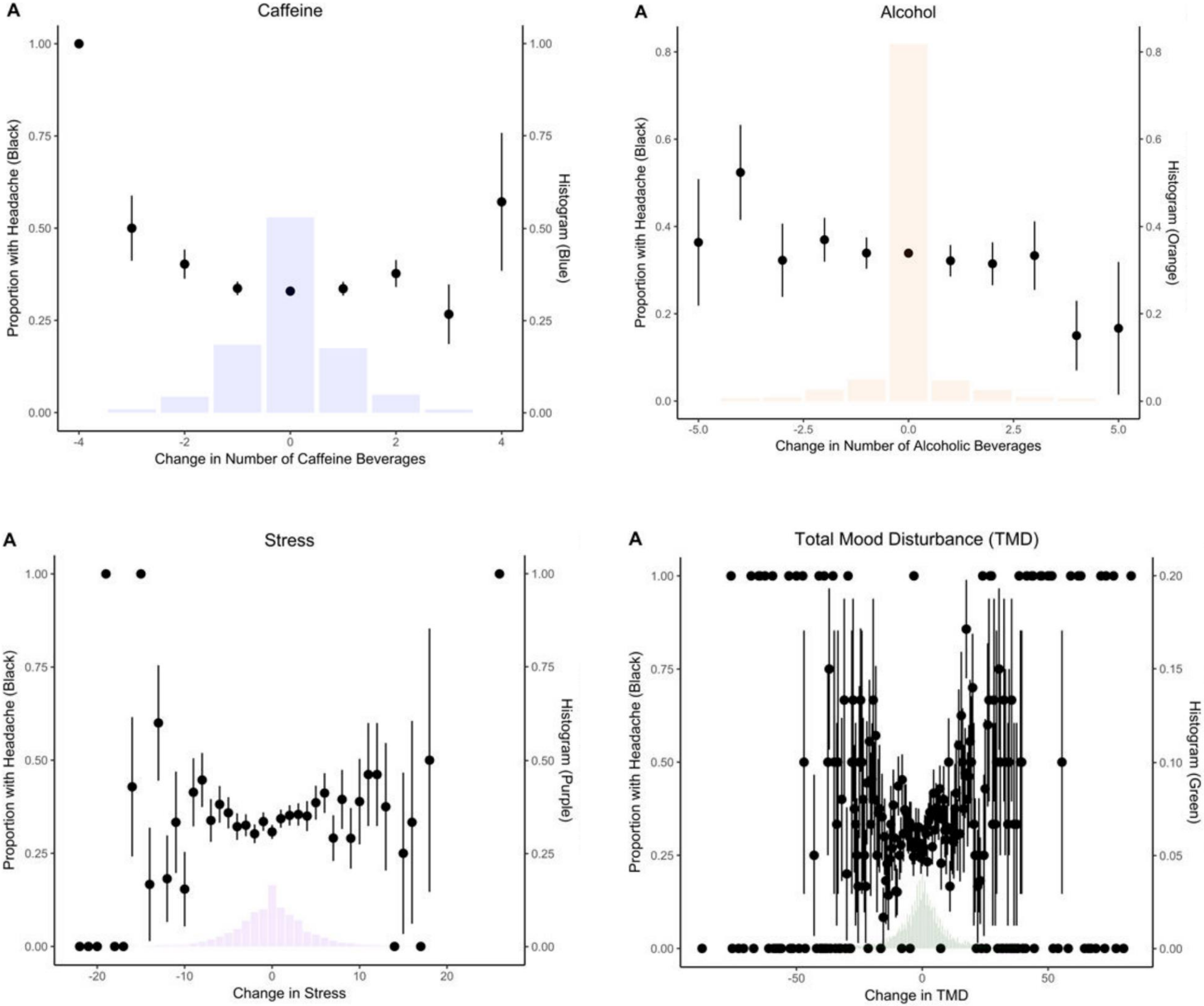
- Information theory can be used to place disparate predictors on the same scale
  - Quantified by the amount of information they contain

- Using this approach, exposure to predictors that are uncommon, or *surprising*, conveys more information about the environment





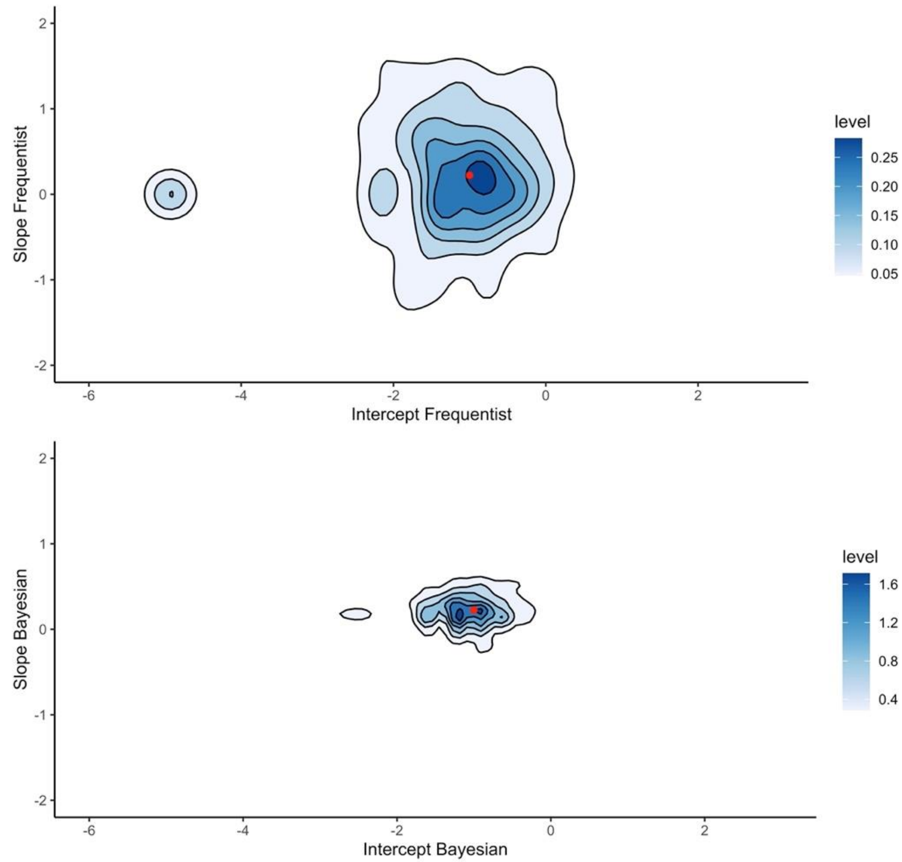
Turner et al, 2020, *Headache*



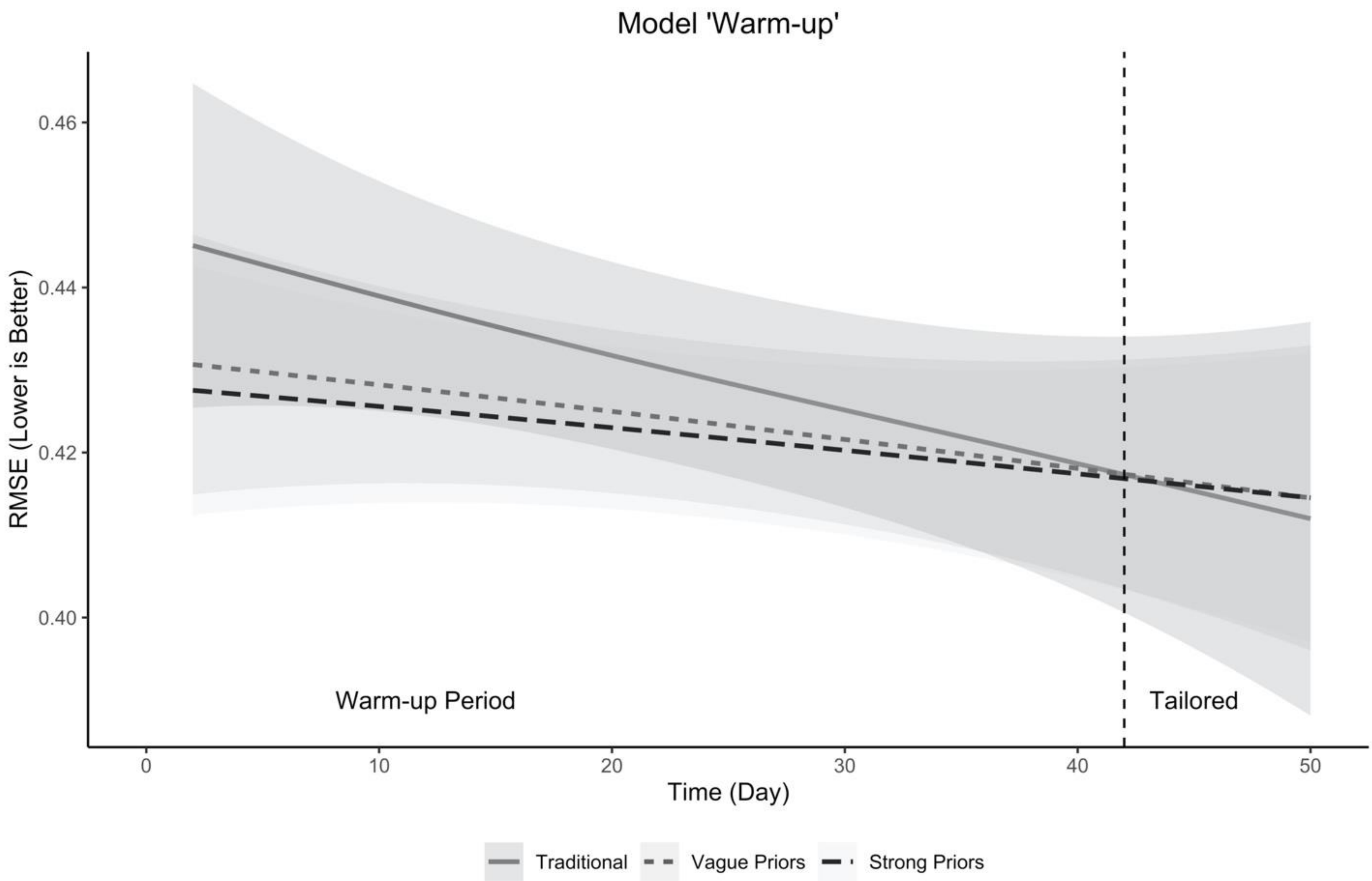
A
Caffeine
Proportion with Headache (Black)
Histogram (Blue)
Change in Number of Caffeine Beverages
1.00
0.75
0.50
0.25
0.00
-4
-2
0
2
4
A
Alcohol
Proportion with Headache (Black)
Histogram (Orange)
Change in Number of Alcoholic Beverages
0.8
0.6
0.4
0.2
0.0
-5.0
-2.5
0.0
2.5
5.0
A
Stress
Proportion with Headache (Black)
Histogram (Purple)
Change in Stress
1.00
0.75
0.50
0.25
0.00
-20
-10
0
10
20
A
Total Mood Disturbance (TMD)
Proportion with Headache (Black)
Histogram (Green)
Change in TMD
1.00
0.75
0.50
0.25
0.00
0.20
0.15
0.10
0.05
0.00
-50
0
50

Continuous updating of individual headache forecasting models using Bayesian methods

- Headache forecasting models must overcome several challenges
  - Heterogeneity in predictor weights
  - Sparsity in data (i.e., few attacks to predict)
  - ‘Warm-up’ time for new models

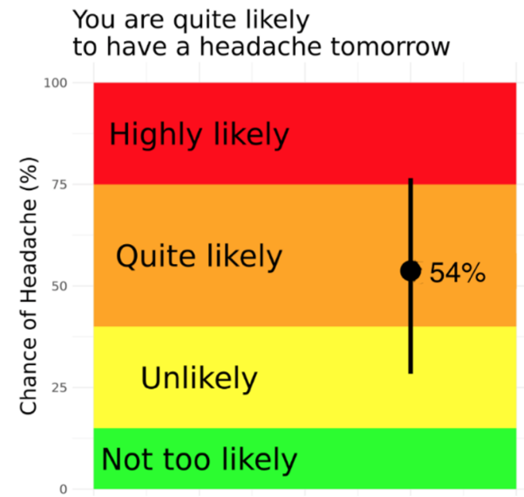


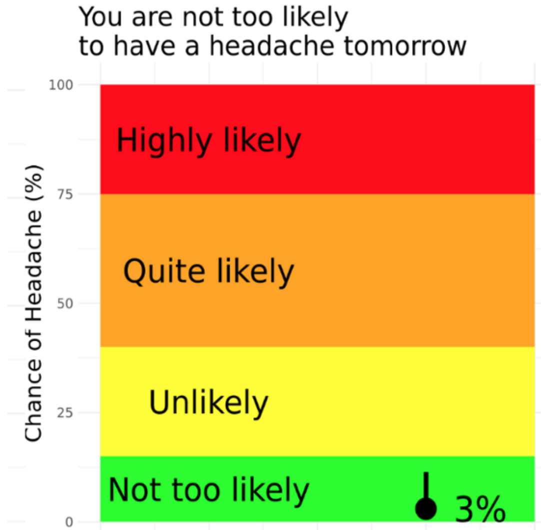
Houle et al., 2021, *Headache*


Model 'Warm-up'
RMSE (Lower is Better)
0.46
0.44
0.42
0.40
Warm-up Period
Tailored
0
10
20
30
40
50
Time (Day)
Traditional
Vague Priors
Strong Priors

Headache Prediction Study II (HAPRED-II)

- 2015 to 2021 (enrollment just closed)
- N = 239 individuals
  - 23,559 forecasts
- Examine
  - Accuracy of predictions
  - Calibration of predictions
  - Reactivity to predictions





Summary

- We have developed and refined several prediction models for migraine (HAPRED-I ,HAPRED-II)
  - Based on Bayesian methods
  - ‘Warm-up’ fast
  - Well calibrated
  - Only modest discrimination
  - Delivery forecasts every 24 hours
  - Consider contextual, non-linear

- Future developments
  - HAPRED-III
  - Self-guided treatment facilitation
  - RCTs (e.g., preemptive treatment)

Contributors

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Thanks!

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